A. Median Maximization

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: binary search, greedy, math | https://codeforces.com/problemset/problem/1566/A

You are given two positive integers n and s. Find the maximum possible median of an array of n **non-negative** integers (not necessarily distinct), such that the sum of its elements is equal to s.

A **median** of an array of integers of length m is the number standing on the $\lceil m/2 \rceil$-th (rounding up) position in the non-decreasing ordering of its elements. Positions are numbered starting from 1. For example, a median of the array [20,40,20,50,50,30] is the $\lceil m/2 \rceil$-th element of [20,20,30,40,50,50], so it is 30. There exist other definitions of the median, but in this problem we use the described definition.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Description of the test cases follows.

Each test case contains a single line with two integers n and s ($1 \le n, s \le 10^9$) — the length of the array and the required sum of the elements.

Output

For each test case print a single integer — the maximum possible median.

Example

Input

```
8
1 5
2 5
3 5
2 1
7 17
4 14
1 1000000000
1000000000 1
```

Output

```
5
2
2
0
4
4
1000000000
0
```

Note

Possible arrays for the first three test cases (in each array the median is underlined):

- In the first test case [underline5]

- In the second test case [underline2, 3]
- In the third test case [1, underline2, 2]